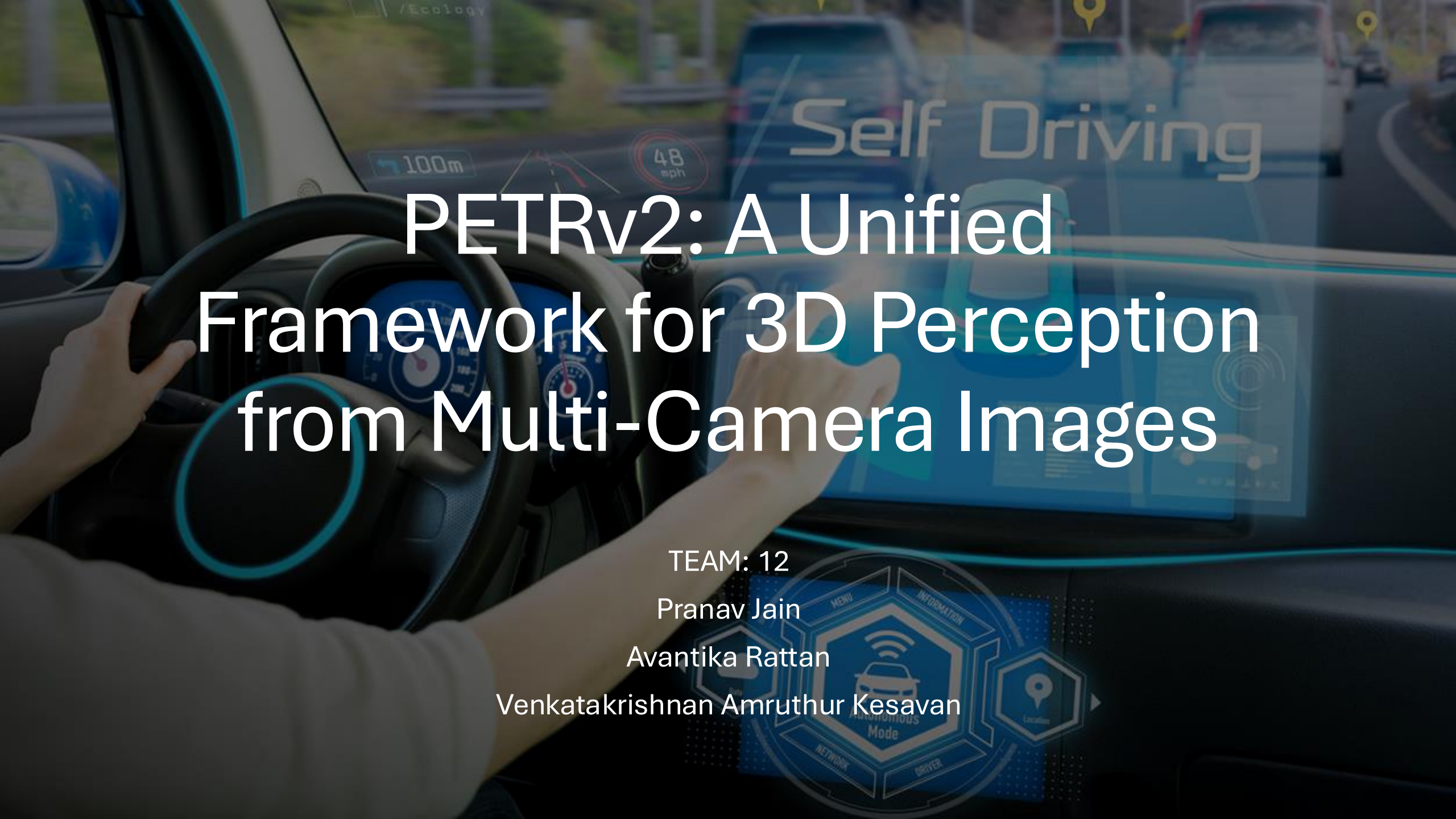




PETRv2: A Unified Framework for 3D Perception from Multi-Camera Images

TEAM: 12

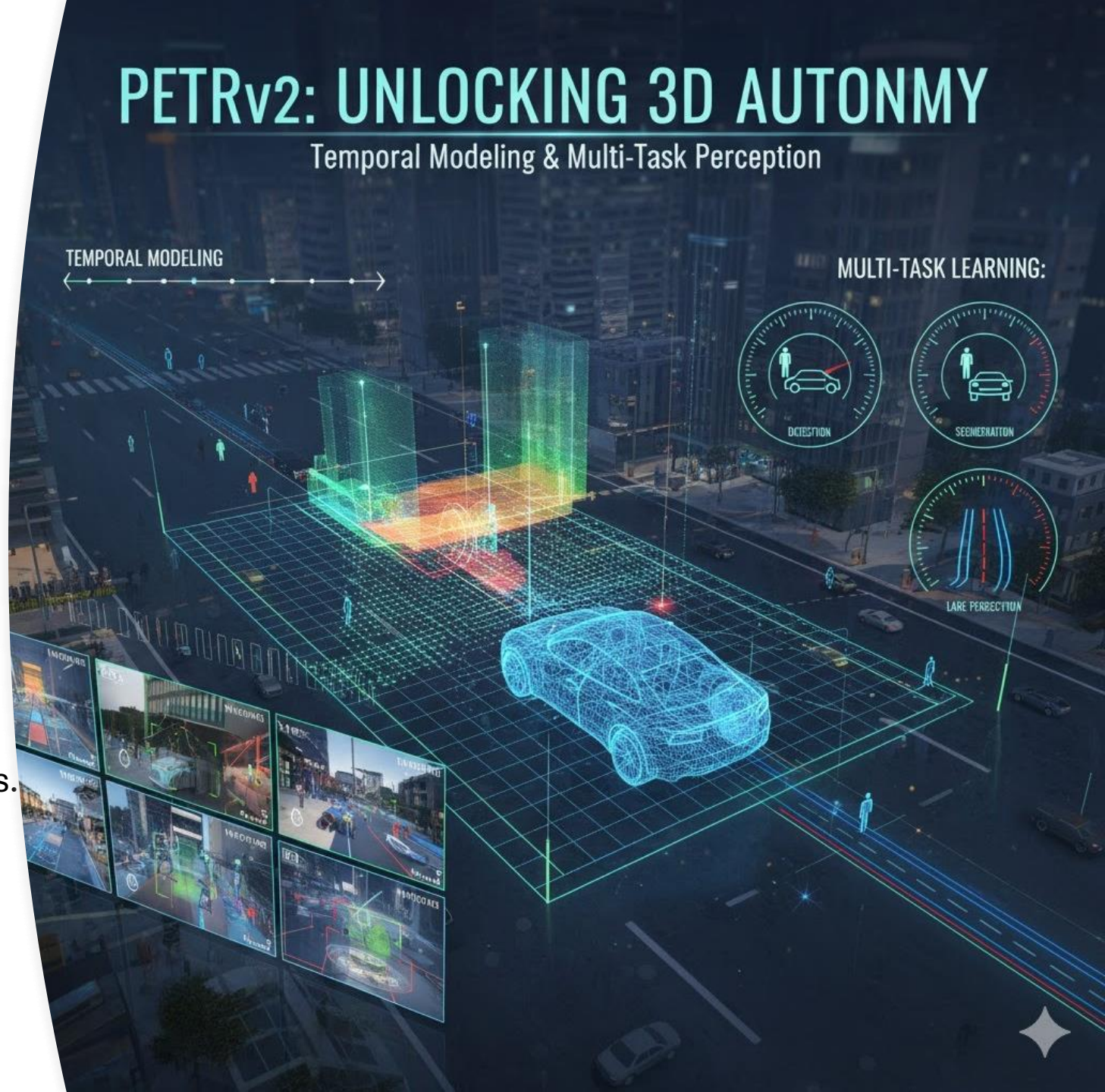
Pranav Jain

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Motivation: Why PETRv2?

- Real driving scenes are dynamics-objects move, lane curves, lighting changes.
- We need a model that:
 - Understands motion by aligning 3D points across frames.
 - Handles multiple tasks (object detection, BEV segmentation, 3D lane detection).
 - Works efficiently using only camera inputs.



From PETR to PETRv2

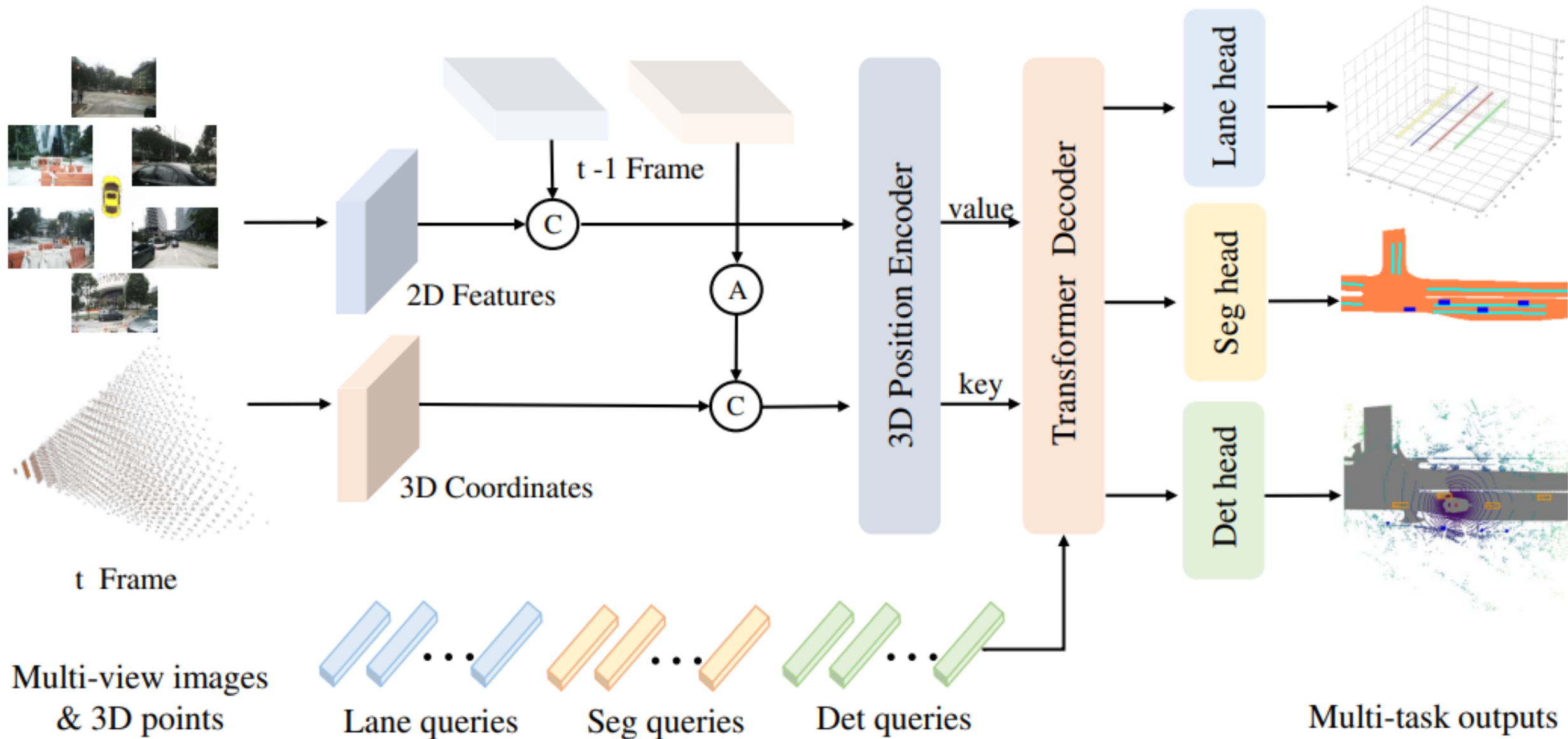
PETR

- Built on DETR transformer framework
- Converts 2D image features → 3D position-aware features
- Injects spatial information (pixel-to-world mapping)
- **Limitations:**
 - Treats each frame independently (no temporal awareness)
 - Single-task only – 3D detection, not segmentation or lane understanding

PETRv2

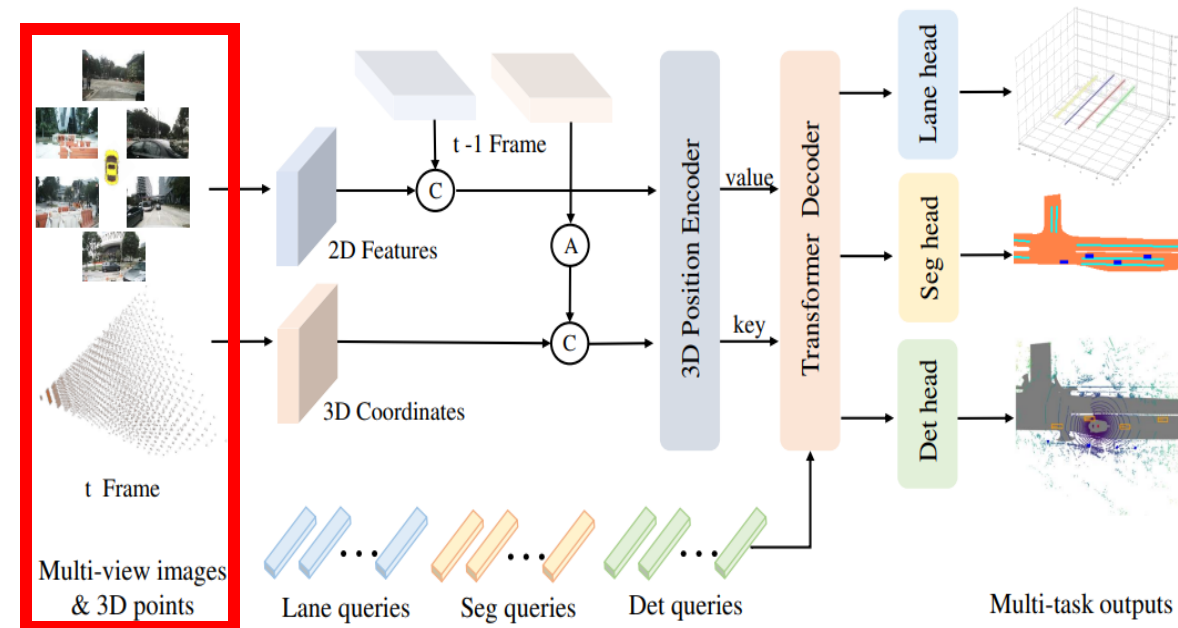
- Adds temporal modeling using ego-pose alignment → tracks motion across frames
- Introduces multi-task transformer with task-specific queries:
 - 3D Object Detection
 - BEV Segmentation
 - 3D Lane Detection
- Unified, end-to-end 3D perception system

Architecture Overview



Input & Feature Extraction:

- 6 multi-camera images (360° coverage)
- Processed by 2D backbone (ResNet / VoVNet) → feature maps

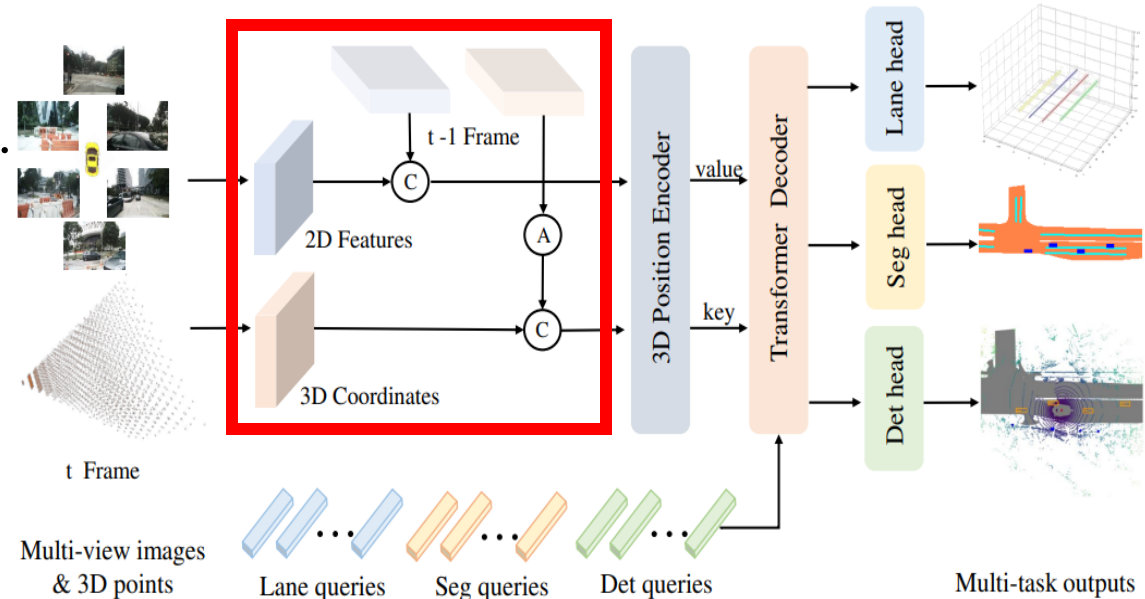


Temporal Alignment via 3D Coordinate Transformation

- Temporal consistency achieved by aligning 3D points across frames.
- Transformation from previous frame (t-1) to current frame (T):

$$\mathbf{P}_i^{l(t)}(t) = \mathbf{T}_{l(t)}^{l(t-1)} \mathbf{P}_i^{l(t-1)}(t-1)$$

- T represents the ego-motion pose transformation
- Alignment happens in 3D space, not 2D feature space.
- Enables motion and velocity estimation for moving objects.



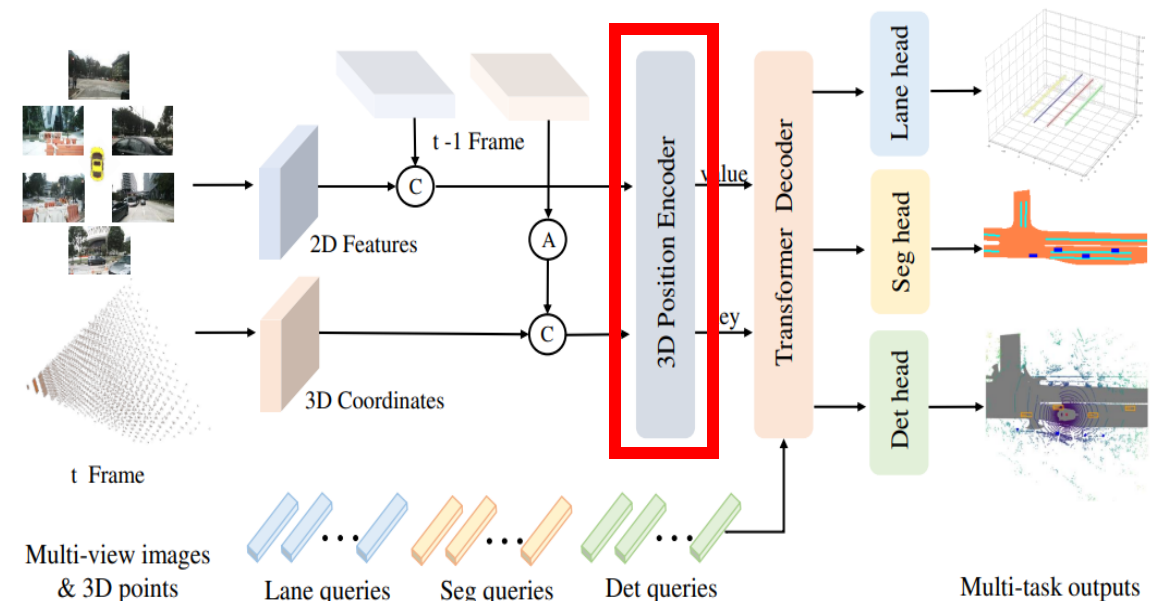
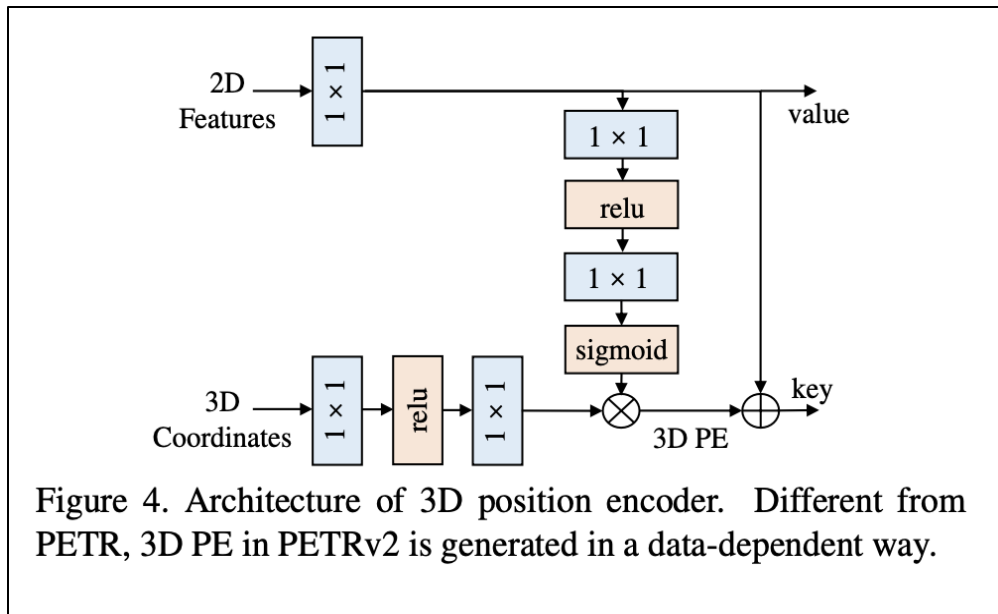
Feature-Guided 3D Position Embedding

- PETR used a static 3D position embedding (data-independent).
- PETRv2 introduces a data-dependent encoder guided by 2D features.

- Mathematical formulation:

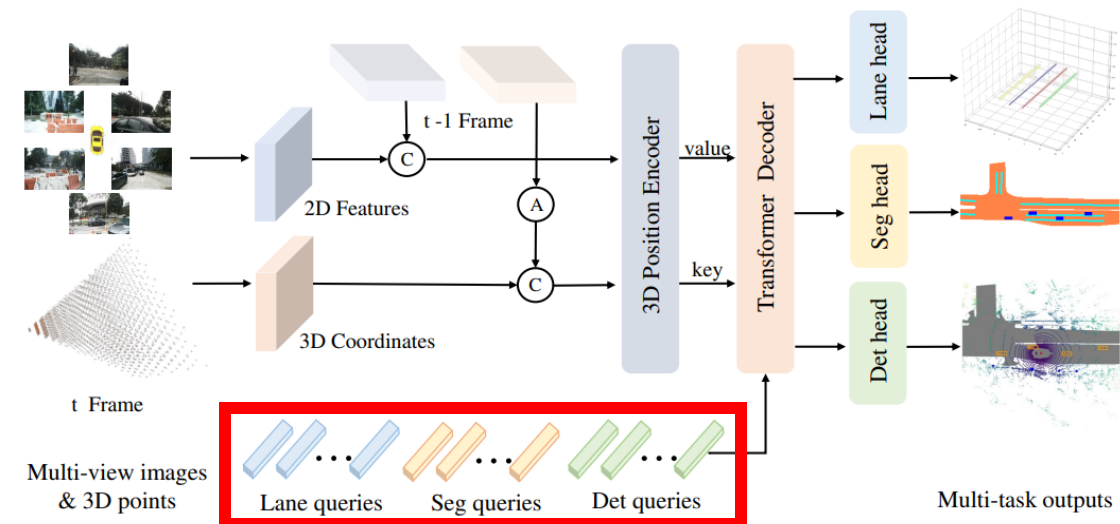
$$\mathbf{PE}_i^{3D}(\mathbf{t}) = \xi(\mathbf{F}_i^{2D}(\mathbf{t})) \times \psi(\mathbf{P}_i^l(\mathbf{t}))$$

- ξ : small MLP + Sigmoid $\rightarrow$ attention weights
- Adds visual context to 3D coordinates $\rightarrow$ better localization & temporal fusion.



Unified 3D Queries

- PETRv2 introduces three task-specific query types and initialized as:
 - **Detection queries:** anchor points in 3D space for object boxes
 - **Segmentation queries:** fixed BEV grid points for map patches
 - **Lane queries:** anchor lane curves for 3D lane geometry
- These initialized queries are then projected onto the feature space using an MLP
- All queries share the same transformer decoder, interacting with temporal 2D and 3D features
- Enables joint learning and shared representation across tasks
- Efficient — uses sparse queries instead of dense BEV maps, reducing computation



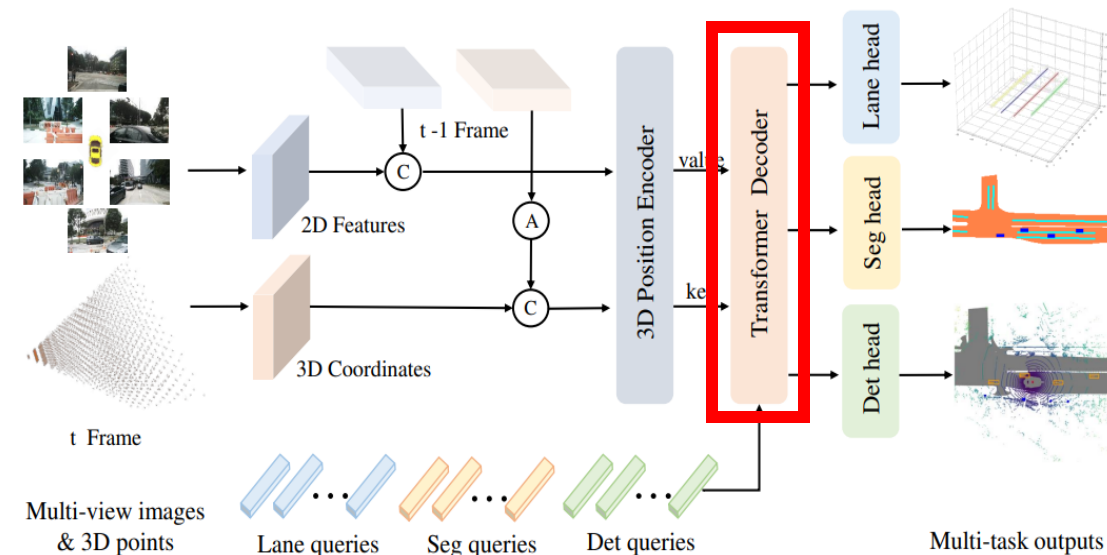
Transformer Decoding

- Decoder takes multi-frame temporal 3D-aware features as keys & values: F^{3D} , F^{2D} .
- Each layer 'l' updates task-specific queries:

$$Q^l = D_l(F^{3D}, F^{2D}, Q^{l-1})$$

At each layer: -

- Step 1: Self-attention = maintains spatial-temporal relationships among queries.
 - Step 2: Cross-attention = queries attend to the 2D & 3D features thus learning relevant info at pixels & patches.
 - Step 3: Feed Forward Network = MLP + ReLu which outputs the layer's updated queries.
-
- Final-layer queries go to the different task heads.



State-of-the-Art Performance on nuScenes

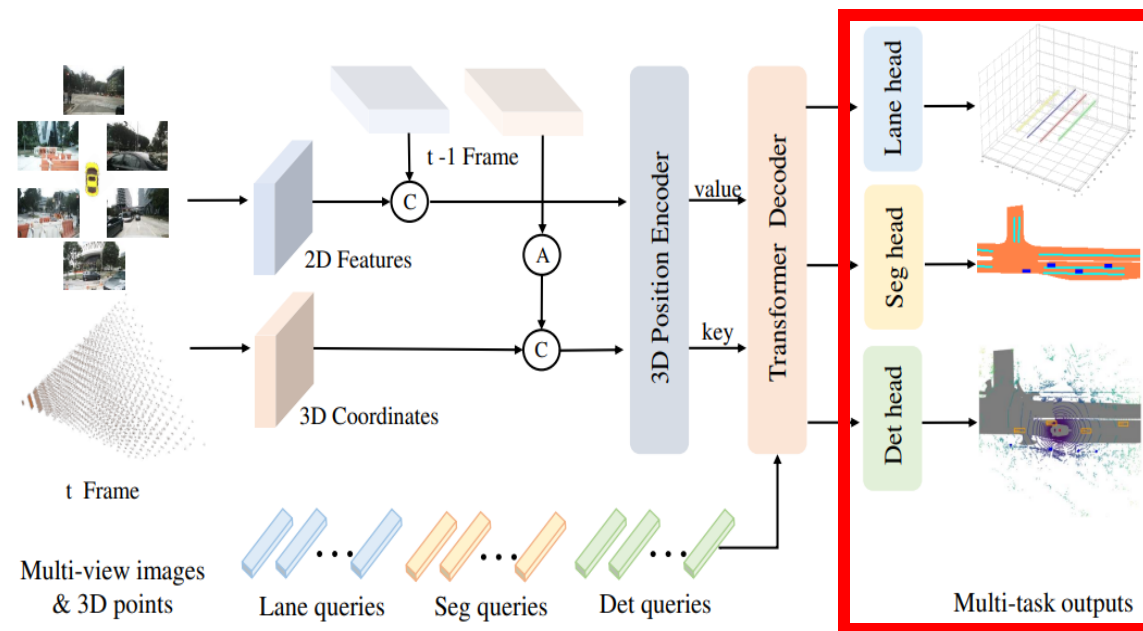
Table 1. Comparison of recent works on the nuScenes val set. The results of FCOS3D and PGD are fine-tuned and tested with test time augmentation. The DETR3D, BEVDet and PETR are trained with CBGS [53]. Inference speed of PETRv2 on a single 3090 GPU, "v" is tested on a single V100 GPU.

Methods	Backbone	Pretrained	Size	NDS↑	mAP↑	mATE↓	mASE↓	mAOE↓	mAVE↓	mAAE↓	FPS
CenterNet [52]	DLA	✗	-	0.328	0.306	0.716	0.264	0.609	1.426	0.658	-
FCOS3D [42]	Res-101-D	✗	1600×900	0.415	0.343	0.725	0.263	0.422	1.292	0.153	1.7(v)
PGD [41]	Res-101-D	✗	1600×900	0.428	0.369	0.683	0.260	0.439	1.268	0.185	1.4(v)
BEVDet [13]	Swin-T	✗	1408×512	0.417	0.349	0.637	0.269	0.490	0.914	0.268	15.6
DETR3D [43]	Res-101-D	✓	1600×900	0.434	0.349	0.716	0.268	0.379	0.842	0.200	6.3
PETR [25]	Res-101-D	✓	1408×512	0.441	0.366	0.717	0.267	0.412	0.834	0.190	5.7
BEVFormer [21]	Res-101-D	✓	1600×900	0.517	0.416	0.673	0.274	0.372	0.394	0.198	3.0
BEVDet4D-T [12]	Swin-T	✗	704×256	0.453	0.323	0.674	0.272	0.503	0.429	0.208	15.5
BEVDet4D-B [12]	Swin-B	✗	1600×640	0.515	0.396	0.619	0.260	0.361	0.399	0.189	1.9
PETRv2	Res-50	✗	704×256	0.456	0.349	0.700	0.275	0.580	0.437	0.187	18.9
PETRv2	Res-101-D	✓	1600×640	0.524	0.421	0.681	0.267	0.357	0.377	0.186	3.5

Table 2. Comparison of recent works on the nuScenes test set. * are trained with external data. ‡ is test time augmentation. "ms" indicates using the resolution of 800 × 320 and 1600 × 640 as the inputs.

Methods	Backbone	NDS↑	mAP↑	mATE↓	mASE↓	mAOE↓	mAVE↓	mAAE↓
CenterNet [52]	DLA	0.400	0.338	0.658	0.255	0.629	1.629	0.142
FCOS3D‡ [42]	Res-101-D	0.428	0.358	0.690	0.249	0.452	1.434	0.124
PGD‡ [41]	Res-101-D	0.448	0.386	0.626	0.245	0.451	1.509	0.127
DD3D*‡ [32]	V2-99	0.477	0.418	0.572	0.249	0.368	1.014	0.124
DETR3D* [43]	V2-99	0.479	0.412	0.641	0.255	0.394	0.845	0.133
BEVDet [13]	Swin-S	0.463	0.398	0.556	0.239	0.414	1.010	0.153
BEVDet* [13]	V2-99	0.488	0.424	0.524	0.242	0.373	0.950	0.148
M ² BEV [46]	X-101	0.474	0.429	0.583	0.254	0.376	1.053	0.190
PETR* [25]	V2-99	0.504	0.441	0.593	0.249	0.383	0.808	0.132
BEVFormer* [21]	V2-99	0.569	0.481	0.582	0.256	0.375	0.378	0.126
BEVDet4D‡ [12]	Swin-B	0.569	0.451	0.511	0.241	0.386	0.301	0.121
PETRv2*	V2-99	0.592	0.506	0.536	0.243	0.359	0.349	0.120
PETRv2* ms	V2-99	0.601	0.519	0.527	0.242	0.358	0.341	0.118

- Outperforms BEVFormer (+2.5 % NDS) & BEVDet4D
- **PETRv2 Vs PETR** → +9.7%NDS, +7.8%mAP
- Temporal fusion significantly improves motion estimation
- Efficient camera-only framework (no LiDAR)



Limitations & Future Work

- **Limitations:**

- Depends on accurate camera calibration & pose estimation → errors affect temporal alignment
- Challenged by adverse weather, low light, or missing camera views
- Transformer-based models are computationally heavy → limits real-time use on embedded hardware

- **Future Directions:**

- Develop lighter backbone networks for faster inference
- Improve robustness to noise, occlusion, and environmental variation
- Explore multi-modal fusion (camera + radar + LiDAR) for stronger perception

THANK YOU



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Appendix A: Table Comparing with Other Works

Table 1. Comparison of recent works on the nuScenes val set. The results of FCOS3D and PGD are fine-tuned and tested with test time augmentation. The DETR3D, BEVDet and PETR are trained with CBGS [53]. Inference speed of PETRv2 on a single 3090 GPU, "v" is tested on a single V100 GPU.

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Appendix B: Table Comparing with Other Works

Table 3. Results on Waymo v1.4 dataset with LET-mAPL metrics. These model are trained on ResNet101 backbone.

Methods	LET-mAPL	LET-mAP	LET-mAPH
BEVFormer-MS	35.4	51.1	47.1
MV-FCOS3D++	35.1	48.0	45.5
PETrv2	36.6	51.9	48.0

Table 4. Comparison of recent BEV segmentation works on the nuScenes val set. * are trained with external data. The performance of M²BEV is reported with X-101 [47] backbone.

Methods	Backbone	Drive	Lane	Vehicle
Lift-Splat [34]	Res-101	0.729	0.200	0.321
FIERY [11]	Res-101	-	-	0.382
M ² BEV [46]	X-101	0.759	0.380	-
BEVFormer [21]	Res-101	0.801	0.257	0.448
PETrv2	Res-101	0.833	0.448	0.434
PETrv2*	V2-99	0.856	0.490	0.463

Table 5. Comparison of recent 3D lane detection works on OpenLane benchmark. PETrv2-V and PETrv2-E are our method with VoV-99 [18] and EfficientNet [38] backbones. PersFormer is reported with EfficientNet backbone. ‡ denotes projecting 2D lane results from CondLaneNet [23] to BEV using IPM.

Methods	F-score(%)↑	X-near↓	X-far↓	Z-near↓	Z-far↓
3D-LaneNet [8]	44.1	0.479	0.572	0.367	0.443
Gen-LaneNet [9]	32.3	0.591	0.684	0.411	0.521
Cond-IPM‡	36.6	0.563	1.080	0.421	0.892
PersFormer [5]	50.5	0.485	0.553	0.364	0.431
PETrv2-E	51.9	0.493	0.643	0.322	0.463
PETrv2-V	61.2	0.400	0.573	0.265	0.413

Appendix C: Ablation Study and Robustness analysis

Table 6. The impact of 3D coordinates alignment (CA) and temporal information (Tem).

CA	Tem	NDS↑	mAP↑	mATE↓	mAOE↓	mAVE↓
		0.437	0.372	0.757	0.479	0.782
	✓	0.426	0.329	0.819	0.500	0.593
✓	✓	0.487	0.396	0.759	0.464	0.423

Table 7. The number of sample points for each anchor lane in 3D lane detection.

Sample Points	F-score(%)↑	X-near↓	X-far↓	Z-near↓	Z-far↓
10	57.8	0.427	0.582	0.293	0.421
400	61.2	0.400	0.573	0.265	0.413

Table 8. Result of different loss weights on the nuScenes val set.

Det	Seg	NDS↑	mAP↑	Drive↑	Lane↑	Vehicle↑
0.0	1.0	-	-	0.836	0.438	0.407
1.0	0.0	0.487	0.396	-	-	-
1.0	1.0	0.488	0.396	0.807	0.388	0.411
1.0	2.0	0.486	0.399	0.814	0.398	0.423

Table 9. Quantitative results on the nuScenes val set with extrinsics noises. The metrics in each cell are mAP[%]. $R_{max} = M$ denotes the maximum angle of three axes is M in degree.

	baseline	$R_{max} = 2$	$R_{max} = 4$	$R_{max} = 6$
PETR	37.9	↓-1.16	↓-3.29	↓-5.08
BEVFormer	41.6	↓-0.71	↓-2.22	↓-3.72
PETRv2	42.1	↓-0.68	↓-2.25	↓-3.66

Table 10. The performance of our method with pose transformation errors. T_{max} denotes maximum translation distance (m).

T_{max}	0.0	0.2	0.4	0.6	0.8	1.0
NDS	52.38	52.38	52.33	52.28	52.17	52.04

Table 11. The performance impact (on mAP metric) of camera time delay. Here, the time delay unit $T \approx 0.083s$.

Time delay	0	T	2T	3T
PETRv2	40.1	36.9 (↓3.2)	34.0 (↓6.1)	30.9 (↓9.2)